Eczematous Dermatoses

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Eczematous dermatoses, that are everyday observations in clinical practice, are induced by various factors of an exogenous and endogenous nature, some better known than others, that can act individually or, more often, in combination.

The nomenclature of these clinical forms seems to be fairly controversial and has not been standardized, both because it is not possible to produce a sufficiently satisfactory classification based on aetiologic or pathogenic criteria and because one or more clinical pictures of eczema can be present in the same patient at the same time or consecutively. The term eczema (from the Greek éczema, in turn stemming from eczéo = to boil) denominates a clinical and histological pattern of skin inflammation that characterizes various dermatoses with different aetiologies. From the clinical standpoint, eczematous dermatoses involve variable pruritus and an ample range of variable symptoms depending on the clinical phase, such as erythema, dryness, exudation, excoriations, scaling, blistering, hyperkeratosis, lichenification, and fissuring. Histologically, at the epidermic level there is spongiosis with various degrees of acanthosis and hyperkeratosis, accompanied at the dermal level by a lymphohistiocytic infiltrate [1].

A helpful classification of the various forms of eczema is to subdivide them into exogenous and endogenous (Table 2.1). However, it must be borne in mind that this is not a clearcut subdivision since in exogenous forms the external triggering factors are associated with an inherited predisposition, while in endogenous forms both internal and external precipitating factors contribute to induce the disease.

Eczematous dermatoses account for a large proportion of all skin diseases, although studies of the relative prevalence have mostly been focused on atopic dermatitis, whereas few have addressed other types of eczema. A study conducted in the USA in a sample of over 20,000 people showed that nearly one-third of them had some significant skin disease. The prevalence of eczemas was 18 $^{0}/_{00,}$ seven of whom had atopic dermatitis, while dyshidrotic eczema and nummular eczema each accounted for about 2 $^{0}/_{00}$ [2]. Consultations for eczematous dermatoses are dealt with above all in primary care. In addition, various cases are referred to hospital, among which eczema forms accounted between 17% [3] and about one-third of all new

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Table 2.1 Classification of principal eczematous dermatoses

Exogenous eczema
Irritant contact dermatitis
Allergic contact dermatitis
Microbial eczema
Dermatophytide
Endogenous eczema
Atopic eczema
Nummular eczema
Pompholyx
Asteatotic eczema
Gravitational eczema
Pityriasis alba
Juvenile plantar dermatosis
Seborrhoeic dermatitis

dermatological cases [4]. In an analysis conducted between 1982 and 1990 in a national tertiary referral centre in Singapore, eczemas accounted for 34% of all new dermatology cases; 66.3% were classified as endogenous while 13.7% were contact dermatitis [5].

Some forms of eczema are more common in infants and young children, like atopic dermatitis, whereas nummular eczema and pompholyx are less frequent in this age group. Other specific patterns are almost exclusive to children, such as juvenile plantar dermatosis. Nummular eczema occurs particularly in elderly males, as also does asteatotic eczema.

The histopathological features of eczema vary according to the clinical phase: acute, subacute or chronic [6, 7]. In the acute phase spongiosis is prevalent. This is an intraepidermic intercellular edema that consequently results in stretching, rupture of intercellular attachments, and the formation of blisters. Lymphocytic exocytosis is present in the epidermis. The increased intraepidermal mitotic activity leads to acanthosis. The intercellular edema is most evident in the mid-epidermal region, while some intracellular vacuolation can also be observed. In the subacute phase, spongiosis and blistering decline while acanthosis increases, together with a parakeratotic horny layer. In the chronic phase, hyperkeratosis gradually replaces the parakeratosis. Acanthosis is more evident, whereas the edema and exocytosis are notably reduced. The inflammatory infiltrate also increases, especially of mononuclear cells at the upper dermis level, although polymorphs and eosinophils may be present in very acute forms. The trauma of scratching causes superficial erosions. Some degree of lichenification is always present in chronic forms, sometimes together with hyperkeratosis and papillomatosis.

2.1 Nummular Eczema

Nummular eczema, also known as discoid eczema, is a clinical entity characterized by coin-shaped or oval lesions with well-defined borders [1]. It was first described by Devergie [8] in 1857 but clinical aspects of the condition had already been roughly outlined by Rayer in 1845 [9]. According to Chipman [10], other historical clinical descriptions, such as Ormsby orbicular eczema, Brocq neurotic eczema, and Pollitzer recurrent eczematoid affection, are similar and actually refer to the same disease, namely nummular eczema, under different eponyms. Histologically, nummular eczema lacks specific features. For this very reason, as well as for its morphological appearance, there is often overlapping with other eczema variants.

2.1.1 Etiology

The condition is not a precise etiological entity and as a matter of fact, a number of agents acting individually or in a combined fashion have been assessed and suggested as the likely etiological factors. The most frequently quoted are nutritional [11, 12], infective [11, 13–15], and emotional [16, 17] etiologies, together with excessive alcohol intake [18] and dry skin, particularly in the elderly [19–21]. In some patients, nummular eczema has also exceptionally been induced by methyldopa [22], gold [23], as well as peginterferon alpha-2b and ribavirin [24].

Contact irritation and contact sensitivity are uncommonly reported as etiological causes. In particular, anecdotal reports have described

nummular eczema as a consequence of hypersensitivity to aloe [25], ethylenediamine hydrochloride [26], depilatory creams [27], mercury [28], soluble oils [29], scabies treatment [30], and fragrances [31]. Relatively large observations, focused on the incidence of contact allergy in nummular eczema patients, are scarce in number [32–37].

As regards the correlation with age, nummular eczema is predominantly an adult disorder, showing a peak between the ages of 20 and 50 years, although it is also observable, albeit at a low incidence, even in pediatric ages [31, 37–42]. It occurs more often in females but then becomes prevalent in males with advancing age.

2.1.2 Clinical Features

Coin-shaped erythemato-vesicular patches form rapidly, due to the confluence of minute papules and papulovesicles, that can also develop as single lesions on the trunk and limbs, together with such patches (Figs. 2.1, 2.2, 2.3 and 2.4). In the acute phase, the patches are intensely erythematous and very itchy. Then they evolve to a desquamation stage with a peripheral extension and possible central resolution, thus giving rise to ring-shaped lesions (Fig. 2.5). During regression, they appear dry and scaly. The patches vary in size from 1 to 5 cm or more in diameter, are generally few, but may be scattered with a bilateral, symmetric pattern. The sites most commonly affected are the arms and trunk.

In young subjects the back of the hands and forearms can be involved (discoid eczema of the hands and forearms) as a form of irritant occupational dermatitis. Women with this clinical variant often have a history of atopy.

The most common clinical variant, that affects the limbs and trunk, can also be observed in elderly people, often with dry skin, where the complaint is exacerbated by low humidity and central heating. The patches appear firstly on the legs and then extend to the trunk and arms. The complaint has a chronic, recurrent course and worsens in the winter months; it may last for a year or more [43]. It is important to exclude an etiological role of irritant or sensitizing chemical agents.

Histopathology reveals typical acute or subacute eczematous findings, with intercellular edema and spongiotic vesicles in the epidermis, that also shows acanthosis with hyperkeratosis and parakeratosis foci. In the superficial derma, there is evident edema and a perivasal lymphomononuclear infiltrate, as well as some eosinophils.



Fig. 2.1 Coin-shaped patches of nummular eczema



Fig. 2.2 Coin-shaped patches of nummular eczema

2.1.3 Diagnosis and Differential Diagnosis

If few, asymmetric, persistent lesions or those with an unusual configuration are present, an exogenous contact dermatitis should be suspected and patch tests performed. In one series, 33% of patients had positive reactions to rubber chemicals, formaldehyde, neomycin, chromium, and nickel [34]. In another study, positive patch tests were detected in 56% of patients, in particular to metals and fragrances [35]. If just one or very few lesions are present, differential diagnosis must be made with tinea corporis, although even in cases of ring-shaped lesions, nummular eczema has more blistering and erythematous margins. Fresh and culture mycological tests provide the definitive diagnosis.

In cases with disseminated lesions various complaints need to be considered. The spots in nummular psoriasis are not vesicular and they are hyperkeratotic. Pre-lymphomatous eruptions mainly affect the trunk and limb joints; the lesions are never blistering or scabbing, but are more infiltrated and have a bizarre morphology. Pityriasis rosea may transiently resemble nummular eczema, but generally presents the herald patch



Fig. 2.3 Coin-shaped patches of nummular eczema (reproduced from Meneghini and Angelini [4])



Fig. 2.4 Coin-shaped patches of nummular eczema

and characteristic oval patches with long axes parallel to the ribs. Asteatotic eczema usually affects the extensor face of the limbs and the lesions are only mildly erythematous, and decidedly dry.

2.1.4 Relationship of Nummular Eczema and Atopy

The relationship of nummular eczema with atopy is fairly controversial. Various studies showed no association between discoid eczema and atopy [11, 12, 43, 44], partly based on the observation of low serum immunoglobulin E levels in the former compared to the latter [45]. Atopic dermatitis, however, can occur with nummular lesions in both children and adults [39, 46, 47]; the incidence is 9% and 12%, respectively [39]. Recent studies indicate that nummular lesions are the most common atypical morphological variant of atopic dermatitis, in pediatric ages as well as in adults. In children it is not uncommon to observe coexisting or alternating eczematous nummular lesions, along with typical skin folds involvement [47]. Even in adult atopic dermatitis, nummular lesions account for about 50% of atypical morphological variants of atopic dermatitis [46]. Nummular eczema, therefore, is an important pattern of presentation of atopic dermatitis, regardless of its association with high serum total IgE levels, the presence of specific IgE and a personal or family history of atopy [48].



Fig. 2.5 Ring-shaped lesions of nummular eczema

Between non atopic and atopic nummular eczema there are several significant differences. Nummular eczema, unlike atopic dermatitis, rarely develops in the first years of life; it commonly occurs around the age of five. Its natural history is different: cutaneous xerosis is missing and the disease does not generally persist after puberty. Nummular lesions, both in pediatric and adult ages, are more exudative, unlike the more usual dry and scaling atopic lesions, and less numerous, asymmetric and irregularly shaped, and resolve with hyperchromic or hypochromic outcomes (especially in atopics) [31, 41, 49, 50]. The most frequently affected sites are the extensor face of the limbs, while the flexor surface and the face are usually spared. When nummular eczema is etiologically linked to infections, alcohol abuse or exogenous chemical agents, it resolves quickly with specific treatment and prevention, and does not recur.

2.1.5 Treatment

The topical treatment is that of eczema in general, according to the various clinical phases; emollients and topical corticosteroids are useful. In cases localized on the hands, it is important to avoid contact with irritants and sensitizers.

Low environmental humidity conditions need to be corrected, including direct exposure to convection heating systems. Systemic anti-histamines relieve the pruritus. In severe exudative forms a course of broad-spectrum systemic antibiotics, such as oxytetracycline or erythromycin, may be useful, also for prophylactic purposes. In severe refractory cases, potentially useful treatments include topical immunosuppressants, oral steroids and immunosuppressants, PUVA and broad- or narrow-band UVB.

2.1.6 Personal Observations

A retrospective epidemiologic and allergologic study conducted in 29,323 consecutive patients patch-tested from January 1982 to December 2009 for eczematous dermatitis of various types elicited 1022 (3.5%) subjects with nummular eczema [33, 37]. The cases of nummular eczema were diagnosed following standardized clinical-morphological criteria: the presence of single or often multiple coin-shaped or oval

patches, with either an oozing crusted surface in acute lesions, or dry, scaly, and lichenified aspects in chronic forms. Moreover, an essential criterion was a clearly demarcated lesional edge, that is crucial when differentiating nummular from other types of eczema, which feature irregularly shaped lesions and blurred margins.

It is essential to stress the chronic course of the dermatitis in all selected patients. Indeed, cases of nummular eczema with a clear endogenous etiology were not included in our analysis, so clinical forms secondary to intestinal parasitosis, systemic drugs or alcohol intake, and focal bacterial infections were excluded by means of the history, clinical course, laboratory data, and response to specific therapy.

Of the 1022 patients, 589 (57.6%) were males and 433 (42.4%) females. Clinical data and medical history showed that among the 1022 patients, 82 (8%), aged from 5 to 20 years, presented both clinical manifestations of atopic nummular eczema and a personal history of atopy. The remaining 940 subjects (92%) were non atopic. These included 181 subjects (19.3%) over the age of 65 years, with serious xerosis and/or stasis dermatitis of the lower limbs and 759 (80.7%) young adults, ranging from 20 to 64 years old, showing primitive idiopathic nummular lesions. As to age at the time of the disease onset, the peak incidence was found in the third decade, with lower figures in the first decade and from the fifth decade onward. In younger individuals, the affliction occurs more frequently in females, then with increasing age it becomes prevalent in males. The median duration of the symptoms was 8 months (range, 4–60 months), and any part of the body could be affected. Lesions were mainly found on the upper (75.8%) and lower (64.5%) limbs, followed by the trunk (45.6%), dorsum of the hands (35.6%), and face and neck (22.3%).

Of the 1022 subjects with nummular eczema, 1 or more positive reactions to patch testing were observed in 332 cases, 182 females (54.8%) and 150 males (45.2%), accounting for 32.5% of the study population. The allergens most frequently involved were nickel (10.2%), chromium (7.3%), and cobalt (6.1%), followed by paraphenylenediamine (5.8%) and ethylenediamine (3.6%). Topical medicaments and their additives were involved particularly in atopic subjects and in the elderly. Overall, in cases of positive reactions their relevance was registered as high (69.7%). We observed an increasing severity of the disease, with greater numbers of lesions in multiple body areas, in patients resulting positive to patch testing.

As previously underlined, our patients were assessed before patch test execution. Therefore, subjects demonstrating a rapid dermatitis resolution following specific (laboratory-guided) treatment were not patch tested. When considering this, the sensitization observed might be explained as follows: in atopic nummular eczema subjects, similarly to in the elderly with nummular eczema overlapping xerosis and/or stasis eczema, contact allergy can be interpreted as a "complication" of a preexisting dermatitis. This is typically a consequence of reiterated topical application of medicaments and cosmetics. In the remaining individuals, instead, allergic contact dermatitis, clinically presenting with nummular lesions, has likely been "primarily" induced by contact with occupational and/or extraoccupational haptens. In both occurrences, the incidence of contact sensitization we found did not differ much from the known rates reported in the international literature.

Our study, albeit hampered by limitations dependent on its retrospective nature, shows that contact allergy is common in persistent nummular eczema and could also possibly be the primary cause of discoid eczema. Therefore, we advise patch testing in all cases of persistent nummular eczema; indeed, if contact allergy is demonstrated, avoidance of the incriminated allergens, whenever possible, might prove of substantial benefit to the patients.

2.2 Asteatotic Eczema

Also known as "xerotic eczema" or "hiemalis" (from the Latin = winter), or "craquelé", this is a dry form with minimal clinical signs, such as modest erythema and pityriasiform

desquamation (Fig. 2.6). This eczema is usually observed in the elderly.

Although commonly attributed to a reduction in lipids on the skin surface, asteatotic eczema is actually multifactorial, being a cumulative dermatitis due to various factors: a naturally, perennially dry skin (in atopy for example); the further reduction in surface lipids occurring with age; particular diseases, malnutrition or hormonal alterations; increased transpiration in humid environmental conditions; alterations of the corneum and chapping due to industrial or domestic solvents and detergents; low environmental humidity and a dry cold wind. In such conditions the physiological skin defences are impaired [51, 52] and there is hence an increased skin absorption of irritant or sensitizing substances coming in contact with the skin that, in turn, cause further skin damage. Other cofactors may be diuretics in elderly people, zinc deficiency [53], cimetidine [54], and topical corticosteroids [55].



Fig. 2.6 Asteatotic eczema: modest erythema and pityriasiform desquamation

The dermatitis mainly affects children, atopic adults and elderly subjects. The sites most often affected are the legs, forearms and hands; the complaint is more pronounced in the winter months. The skin is dry and mildly scaly and the backs of the hands show a crisscross design of scaly lesions. The fingertips are dry and fissured, with skin like parchment (on pressure, a depression appears and lasts a while). At the level of the legs, the skin appears deeply grooved (hence the term "craquelé" used by French authors). The margins of the patches are often erythematous and slightly raised; frankly eczematous lesions can later develop. The complaint has a chronic course, worsening in winter and improving in summer months. An overlapping contact dermatitis is often observed. The pruritus is intense, particularly when undressing.

Extensive or generalized forms of asteatotic eczema should suggest malignancy [56]. Histopathology shows the features of a mild, subacute eczema with a modest dermal infiltrate. The environmental humidity must be corrected. Wool in direct contact with the skin is poorly tolerated and can cause further irritation. The patient should be advised not to bath too frequently; to apply soothing soaps and emollients all over the body every day, and oily topical creams (based on lanolin or paraffins) on restricted skin areas. If necessary, weak topical corticosteroids in a urea base that increases hydration are useful.

2.3 Gravitational Eczema

The terms "stasis eczema" or "varicose eczema" by which this complaint used to be known are no longer acceptable because varices and stasis are not a requirement for the development of the disease. Instead, it is associated with venous hypertension and increased tissue perfusion. In cases with venous hypertension of the legs there is an increased blood oxygen content and accelerated circulation. This leads to dilatation

of the local capillary bed and hence the escape of fibrinogen from the vessels. The perivasal fibrin layer then acts as a barrier to the spread of oxygen and other nutrient substances and so causes an altered skin vitality [57, 58]. Venous hypertension can induce leukocytes intravenular sequestration and the consequent release of proteolytic enzymes and free radicals, triggering the inflammatory process [59].

The dermatitis initially involves the internal face of the lower third of the legs, and may have an acute or an insidious onset, often the result of a previous deep vein thrombosis. Patients are generally middle-aged or elderly and more often women, probably due to hormonal effects and the tendency to venous thrombosis in pregnancy.

The dermatitis is of erythemato-desquamative type and often exudative, usually accompanied by venous hypertension, edema, purpura, hemosiderosis, ulceration, or small patches of white, atrophic, telangiectatic scarring ("atrophie blanche"). This altered skin condition is very prone to the onset of allergic contact dermatitis, perhaps also because of the large number of HLA-DR + cells in the affected skin [60].

In differential diagnosis it must be borne in mind that various other types of eczema can affect this region, like atopic dermatitis in children and young adults, nummular eczema (usually on the anterior or anterolateral aspect of the lower legs), and asteatotic eczema in the elderly.

The venous hypertension must be controlled by physical means (elastic socks and permanent use of appropriate bandaging) and medicaments. Obese patients need to lose weight. It is important to instruct the patient about the correct posture of the legs, and sometimes bed rest can help. Topical treatment must take into account the increased risk of contact allergy. Mild topical steroids can be used to relieve irritation, but potent steroids must not be used due to the risk of atrophy and hence ulceration. Topical tacrolimus has been reported to be efficacious. Any supervening bacterial infection must be treated with systemic antibiotics and leg soaks with a mildly antiseptic solution. In all cases patch tests are advisable.

2.3.1 Personal Observations

Notoriously, gravitational eczema, with or without ulcerations, can be complicated by contact allergy; the pertinent literature contains sample data, above all about iatrogenic factors in the form of medicaments. In our studies of a total of 1204 patients, contact allergy was demonstrated in 54.9% of the cases [61–63]. The patch test series included 63 substances, among which were numbered medicaments, topical drugs excipients and other haptens. The highest percentage of positive responses was obtained to medicaments, in particular neomycin, benzocaine, sulfamide, promethazine, penicillin, chloramphenicol, and quinolines. Among excipients, the main culprits were parabens, wool alcohols, benzoyl peroxide, Peru balsam, and colophony. Other prevailing haptens were paraphenylenediamine and other clothing dyes (socks, shoes), chromium (shoes), nickel and cobalt (shoe buckles), thiurams (shoes).

The data clearly showed that legs with gravitational eczema are the site most vulnerable to contact allergy, owing to the reduced or impaired integrity of the skin barrier (in cases of ulcerations) [64]. In this disease, the incidence of contact allergy reported in the literature ranges from 40% to 90% (mean about 60%) [65–72]. Thus, preventive measures are essential; they rely mainly on the use of non sensitizing topical products.

2.4 Pityriasis Alba

This is characterized by hypopigmented patches preceded by mild erythema and scaling. It is often a manifestation of atopic dermatitis but does not only affect atopic subjects.

Pityriasis alba is observed above all between the ages of 3 and 16 years in both sexes, and features rounded or oval hypopigmented patches with indistinct margins. Initially the patches are erythematous with fine scaling, then the erythema disappears leaving hypochromic lesions. These are most evident in dark-skinned subjects or in fair-skinned subjects after exposure to the

sun. The lesions range from 0.5 to 2 cm or more in diameter and are localized on the trunk, and in children above all on the cheeks, neck and arms (Figs. 2.7, 2.8, and 2.9). The disorder has a variable course, and the lesions can persist for months; it can also recur. Histology shows modest spongiosis, acanthosis, and parakeratosis. There may also be follicular plugging and sebaceous gland atrophy. The number of active melanocytes is reduced (but not of total melanocytes) and melanosomes are fewer and smaller [73, 74].

An accentuated hypopigmentation must raise the suspicion of vitiligo. Naevus depigmentosus is commonly present at birth or before the age of 3, and is often a single lesion with distinct margins. Nummular eczema in an atopic child features larger lesions that are intensely itchy. In adults, lesions of the trunk must be differentiated from psoriasis. In rare cases, mycosis fungoides presents with hypochromic lesions, and this is difficult to identify in differential diagnosis, even histologically. Treatment, that is not always successful, relies on emollients. The pigmentation will resolve slowly over time. Topical tacrolimus and pimecrolimus seem to be efficacious in cases of facial involvement and inflammation [75].

2.5 Juvenile Plantar Dermatosis

Since 1968 this has been described almost exclusively in children aged between 3 and 14 years, using various synonyms: "peridigital dermatosis" [76], "atopic winter feet" [77], "recurrent juvenile eczema of hands and feet" [78], "forefoot eczema" [38], South Australian feet" [79], "shoe dermatitis in children" [80]. Then, in 1976 the term "juvenile plantar dermatosis" was proposed [81, 82], and it is now known by this name, although this does not



Fig. 2.7 Hypochromic lesions of pityriasis alba



Fig. 2.8 Hypochromic lesions of pityriasis alba

take into account etiological agents, seasonal variations or the geographic distribution of the condition. Although almost always a childhood complaint, there have been anecdotal reports describing the condition in babies in arms [83], and rare cases in adults. Both sexes are affected although there is a slight prevalence in males.

The dermatitis normally appears in children who actively practice sports: the plantar surface of the forefoot appears smooth and shiny, with ragades at the flexory folds of the toes. The support areas of the forefeet and tips of the toes are particularly affected, and the heels may also be involved, but not foot areas that do not come in contact with the ground. The interdigital spaces are normal with no signs of maceration due to fungal infection; in many cases the lateral and medial faces of the first and fifth toes are affected. The dermatitis is bilateral and symetrical. In rare cases of involvement of the hands, the complaint shows the same clinical aspects on the palms and possible fissuration of the fingertips.

The clinical diagnosis is not generally difficult. A family or personal history of atopy should be checked for, although this association is controversial. In doubtful cases it is necessary to perform patch tests to exclude contact allergy to shoe components or sock dyes, or to topical medicaments used to treat the disorder. A mycological test may also be useful. In exceptional cases the histology has been analyzed but conflicting findings have been reported. A block of the sweat ducts may be evident, and hence sweat retention, while other authors have reported a picture of mild chronic eczema or chronic miliaria.

As regards the etiopathogenesis, juvenile plantar dermatitis is linked to the use of synthetic socks and shoes that, unlike natural materials like cotton, wool and leather, are not porous. The natural evaporation of sweat is therefore prevented and the feet are chronically hot and damp and therefore subject to maceration. The latter phenomenon can block the sweat pores causing sweat retention. This concept is supported by the fact that sports are practiced by a very high percentage of affected children, wearing relatively non porous sports shoes, even if the complaint is occasionally observed in subjects wearing open sandals and cotton socks. High proportions of children with plantar dermatitis have a family history (24%) or are personally affected (18%) by atopy, although few of them present active atopic dermatitis lesions at the time of observation [82]. The issue of an association with atopy warrants further study. In any case, it is reasonable to believe that subjects with a history of atopy are more predisposed to this dermatitis, that can develop after even mild exogenous stimuli. In the case of children with a negative history of atopy, instead, very intense physical activity and continual use of gym shoes is required to bring on the complaint. These latter conditions are essential in non atopic subjects. By contrast, children with severe atopic dermatitis do not present plantar dermatitis, presumably because they do not do gymnastic activities.

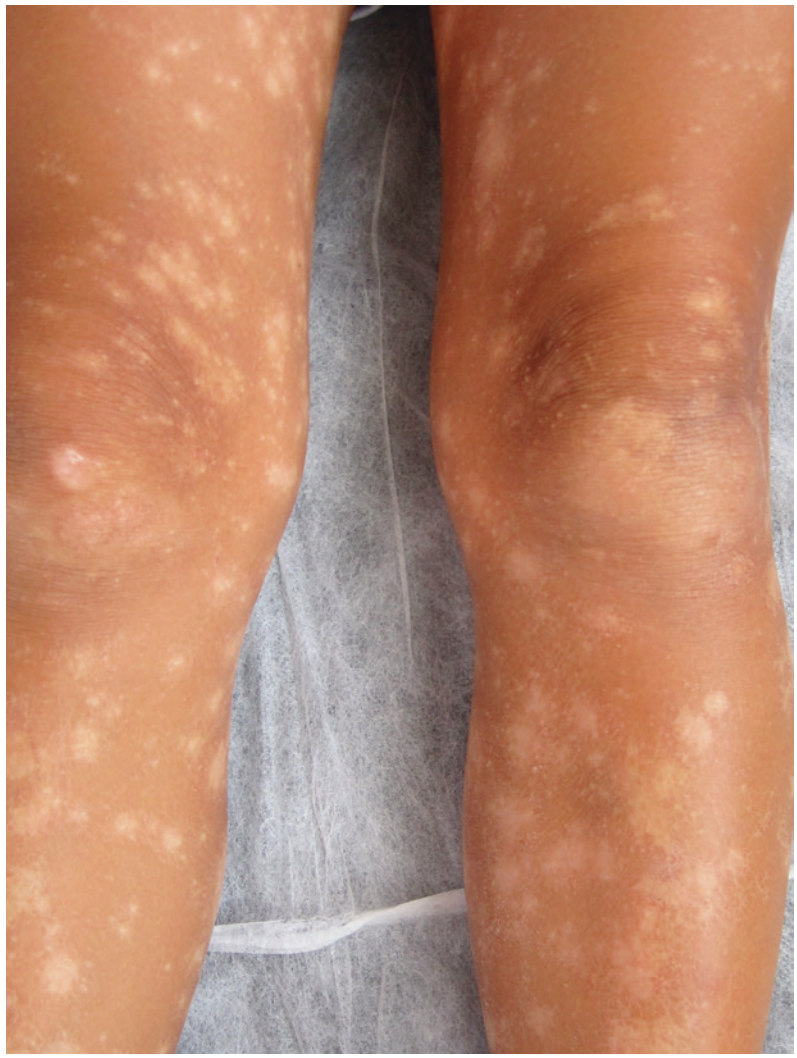


Fig. 2.9 Hypochromic lesions of pityriasis alba

The prognosis is good and the dermatitis generally resolves within about 4 years, either due to adequate treatment or to using appropriate shoes, or else on account of a sort of "hardening" of the plantar skin. Undoubtedly, the use of cotton socks and leather shoes contributes to an evident improvement of the dermatitis, that may resolve spontaneously over time.

Topical treatment is not usually efficacious, but good results can be obtained with topical drugs with a urea, tar or Lassar paste base.

2.6 Pompholyx

Pompholyx (synonyms: vesicular eczema of the palms and soles, dyshidrotic eczema) is a chronic recurrent eruption of vesicles or blisters on non erythematous skin of the palms (cheiropompholyx) and soles (podopompholyx). The term dyshidrotic eczema, that has a supposed connection with sweat gland activity since the condition is worse in hot weather, should be abandoned, as no causal relationship

with the sweat glands or sweating has been demonstrated.

The incidence in the general population is not known, but the range is 5-20% among all cases of eczema of the hands [84]. The etiology is also unknown (Table 2.2). In most cases there are no evident exogenous causes. The role of a hereditary predisposition has not been established. The role of the sweat glands is debated: the dermatitis affects sites with emotionally-induced heavy sweating, and worsens during the warm months, but it is not constantly associated with hyperhidrosis; serial histological sections of the blisters show that they dislodge the sweat ducts that pass between these same blisters [85].

The role of atopy is also controversial: some authors believe there is a relation between the two conditions [86, 87] whereas others regard this as irrelevant [88]. Direct contact with some allergens can sometimes cause an asymmetrical palmar vesicular eruption, rather than the more common frank eczema of the dorsum of the hands [89]. In any case, the possibility of onset of contact allergy superimposed upon primitive pompholyx should be borne in mind [90–92]. The reported high incidence of pompholyx in nickel-sensitized subjects, that could be reproduced also by means of oral challenge with nickel sulfate [93–96], was not confirmed by other authors [97, 98]. The same event has occasionally been observed in subjects allergic to chromium and cobalt [90] and in 3 of 10 subjects allergic to neomycin, after oral challenge with this drug [99].

Table 2.2 Etiological factors in pompholyx

Sweating
Hot weather
Atopy
Stress
Bacterial foci
Mycobacterial foci
Drugs (aspirin, oral contraceptives)
Cigarette smoking
Direct contact allergy
Systemic contact allergy

A mycotic infection of another site, in general the feet, can provoke a palmar dyshidrosic eczema of tinea type (dermatophytide), that may resolve after healing of the starting focus and recur when this reappears. Bacterial foci (bacterides) may act according to the same mechanism. The role of stress is difficult to define: very likely stress and pompholyx have a reciprocal influence. Exceptionally, pompholyx may follow a drug eruption. Aspirin, oral contraceptives and cigarette smoking increase the risk of pompholyx [100].

The onset of pompholyx may occur at any age, although it is more common in young adults and rare before the age of 10 years. The onset is extremely acute, featuring groups of deep, clear blisters resembling sago beans. Erythema is lacking, but pricking, pruritus and heat can precede the eruption. The vesicles can become confluent (Figs. 2.10, 2.11, and 2.12), forming blisters, especially on the feet. The pruritus is intense. The episode will regress spontaneously within 2–3 weeks, although the disease course may feature further episodes in stages. In modest cases only the internal faces of the fingers are affected whereas in more manifest cases the palms and/or soles are symmetrically involved. A unilateral localization is possible, although it should suggest a form of contact allergy. The hands alone are affected in 80% of cases, the hands and feet in 10% and just the feet in 10%. A superimposed infection with pustules and lymphangitis is possible. In chronic, recurrent cases affecting the dorsa of the fingers, the nails are dystrophic, showing irregular horizontal grooves, domed depressions, thickening and dyschromia. In cases with an unknown etiology (the majority) there are frequent recurrences at intervals of about 1 month or more apart: they are more common in the summer.

Histopathology shows the alterations of acute eczema, modified by thickening of the epidermis; in the resolution phase there is evident hyperkeratosis and epidermal shedding (Fig. 2.13).

A very mild form of probable pompholyx is "recurrent focal palmar and plantar peeling" (in the past this was improperly called "keratolysis exfoliativa"). During the summer months small



Fig. 2.10 Pompholyx

areas of whitish superficial scaling appear in the palmo-plantar sites, with no erythema nor evident blistering. This picture is probably not rare but it is often missed due to modest symptoms that do not drive the patient to seek dermatological help.

The diagnosis of pompholyx is easy; differential diagnosis is with tinea manuum, contact dermatitis and pustulous psoriasis. It is important to take into account a possible distant bacterial or mycotic focus. Occasionally, pemphigoid, linear IgA disease, and pemphigoid gestationis can present with large blisters on the palms mimicking pompholyx [101]. The treatment is symptomatic in most cases. In the acute phase applying compresses with a weak antiseptic solution 3–4 times a day (Burow solution, sodium hypochloride or potassium permanganate) can be useful. The content of large blisters must be aspirated with a sterile syringe. In the subacute phase topical steroids can be used, while in the hyperkeratotic phase topical keratolytics are a valid treatment. Any secondary bacterial infection must be treated with systemic antibiotics.

2.6.1 Personal Observations

Pompholyx has a tendency to become a chronic recurrent disease that favours the onset of contact allergy to occupational and non occupational substances [102].



Fig. 2.11 Pompholyx and contact dermatitis

In a first study [91] the disease incidence was assessed in a group of 364 patients (60% males and 40% females) with palmar and/or plantar forms, consecutively observed over 5 years. Patch tests demonstrated major positive reactions to one or more substances in 108 cases (29.6%); among these, 26.8% were positive to topical medicaments or their excipients and 73.2% to other haptens. Among the medicaments, the most common were neomycin, penicillin, sulfamide, and promethazine, and among excipients the most frequent culprits were parabens, Peru balsam, and ethylenediamine. Of the other substances, the highest incidence of positive reactions was to paraphenylenediamine (31.5%), followed by chromium (25%), cobalt (10.2%), mercaptobenzothiazole (9.3%), nickel (6.5%), and *p-tert*-butylphenolformaldehyde resin (2.7%). The degree of positive reactions depended on specific working activities, and on the use of special gloves and shoes. In a subsequent study [92], the above incidence of contact allergy in subjects affected by pompholyx was confirmed, being 31% (14 of 45 subjects). The incriminated agents were the same as in the previous study.

2.7 Seborrhoeic Dermatitis

This is a chronic dermatitis characterized by a peculiar clinical picture in terms of morphology (well defined erythematous patches covered by fatty scales) and sites (areas with many sebaceous glands: the scalp, face and upper trunk). The skin folds can also be affected, although this is not an essential diagnostic criterion. The incidence is 3–5% in the general population, being more prevalent in young adults and in subjects with dandruff and visible scaling of the scalp. The incidence is very high in subjects with the initial stages of HIV infection: a study of patients with a normal helper T-cells count and delayed hypersensitivity revealed an incidence of 36% [103].

2.7.1 Etiology

It is generally agreed that yeast of the genus *Malassezia*, that is increased in the scaly epidermis in dandruff and seborrhoeic dermatitis conditions, causes the condition [104]. The mechanism of action whereby *Malassezia* spp. induces the disease is not clear, and the main evidence of an etiological role remains the positive response to specific treatment against this genus [104].

The activity of the sebaceous glands at birth, linked to stimulation by maternal androgens, ceases when this stimulation ends; the sebaceous glands then remain inactive for 10–12 years. In fact, infant seborrhoeic dermatitis is confined to the first months of life, although it is not yet certain that this is the same condition that appears in adolescence and adulthood. This latter condition is rare before puberty, reaches a peak between 18 and 40 years, and is occasionally



Fig. 2.12 Pompholyx and contact dermatitis

observed in the elderly. Seborrhoeic dermatitis is more common in men than women.

The role of seborrhoea in the pathogenic mechanism of seborrhoeic dermatitis is doubtful, although the maturation of the sebaceous glands can be a factor favouring the disease onset. Many young adults with the disease have oily skin, but frontal sebaceous excretion measured in subjects with classic seborrhoeic dermatitis appears to be normal in males and significantly reduced in females. On this basis, it has been suggested that the term "dermatitis of the sebaceous areas" would be more pertinent than seborrhoeic dermatitis. The belief that seborrhoeic dermatitis is more common in subjects with acne has no scientific basis.

No qualitative abnormalities in the composition of the sebum have been demonstrated. Seborrhoeic skin is highly susceptible to bacterial agents and physical and chemical stimuli,

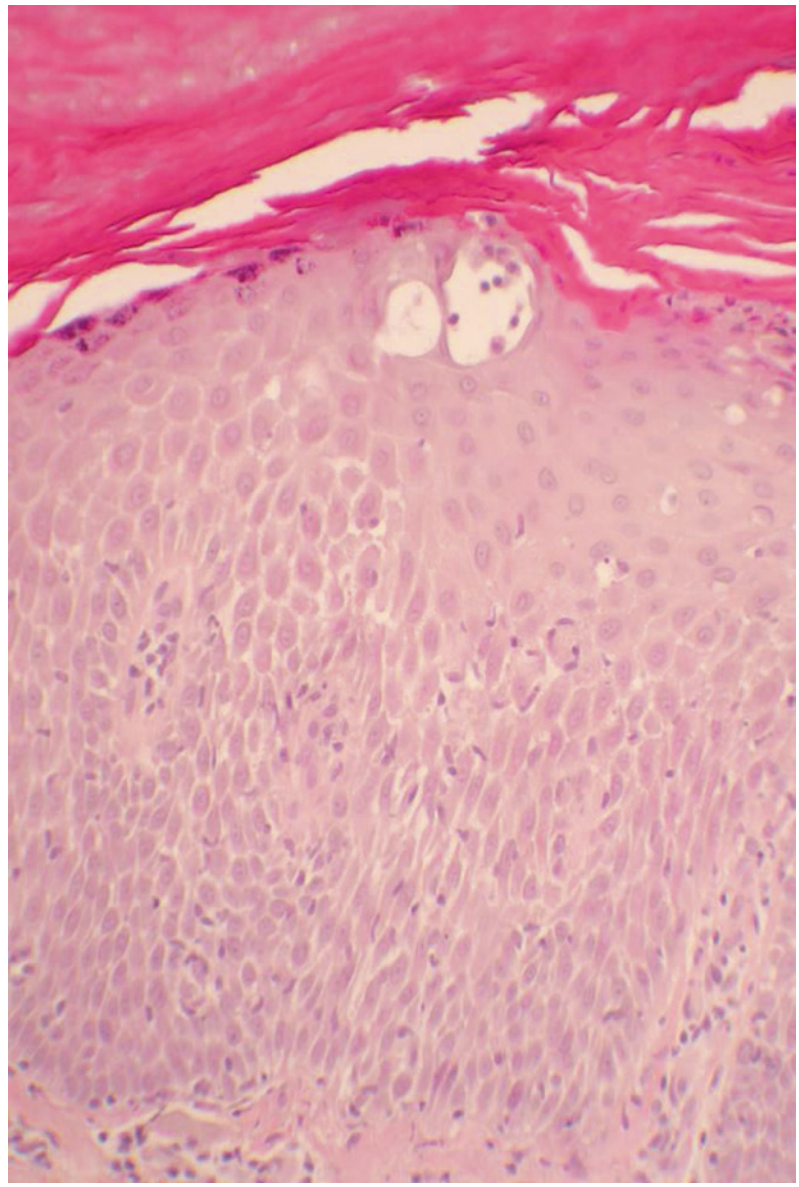


Fig. 2.13 Histopathology of pompholyx: confluent intraepidermal vesicles

that can therefore bring on infections and contact dermatitis.

2.7.2 Histopathology

The histopathology demonstrates hyperkeratosis with parakeratotic foci, moderate acanthosis and modest spongiosis. A mild chronic inflammatory infiltrate is present in the derma. Ultrastructural studies have shown findings more similar to those of nummular eczema than to irritant contact dermatitis or allergic contact dermatitis. In patients with AIDS there is a greater follicular involvement, with numerous plasma cells, neutrophils and intraepidermal nuclear dust, as well as a remarkable quantity of yeasts.

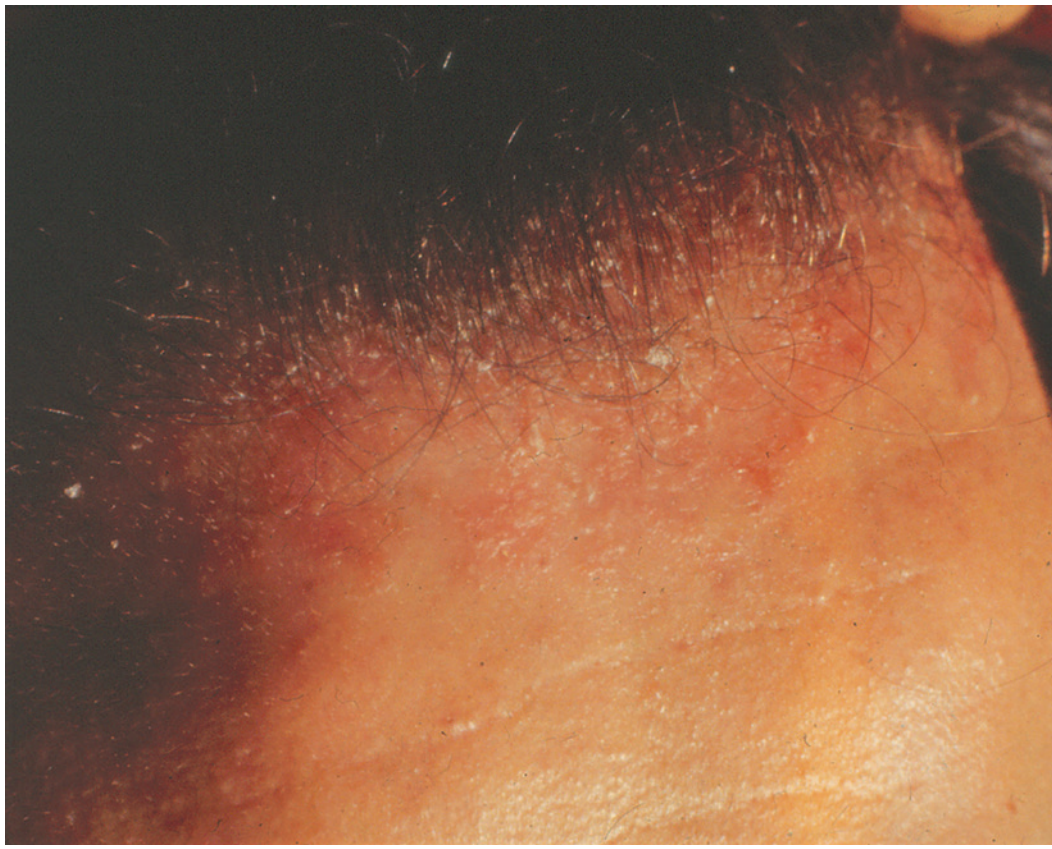


Fig. 2.14 Seborrhoeic crown beyond the hair line

2.7.3 Clinical Aspects

Seborrhoeic dermatitis presents various clinical variants (Table 2.3), that usually start in hairy zones, involving the scalp, face, presternal and interscapular regions and the skin folds. The lesions are reddish-yellow and covered by oily scales. The first manifestation is usually simple dandruff of the scalp, followed by a perifollicular erythema stage and the gradual formation of patches with distinct margins, consisting isolated or confluent lesions that will spread over much of the scalp and extend beyond the hair line (a "seborrhoeic crown") (Fig. 2.14). In chronic cases this form can be accompanied by hair loss, that may resolve when the inflammation dies down. It has not been established whether seborrhoeic dermatitis may accelerate the appearance of androgenic alopecia.

The retroauricular regions are erythematous with abundant oily scaling and possibly ragades and scabs in the folds. From there the dermatitis can extend to the ears, periauricular regions and sides of the neck.

Table 2.3 Clinical variants of seborrhoeic dermatitis

Childhood
Scalp (cradle cap)
Trunk (flexures, napkin area)
Leiner's disease
Adults
Scalp (dandruff, inflammatory form)
Face
Trunk
Petaloid
Pityriasiform
Follicular
Flexural
Generalized

On the face, the disease characteristically affects the internal eyebrow area, the glabella and the naso-labial grooves (Fig. 2.15). Blepharitis is frequent: the eyelid margins (Fig. 2.16) are erythematous and covered with small whitish scales. In this site, an alternating course is very common, with recurrence in times of stress. Exposure to light can aggravate the problem but tanning will improve it. A modest form of seborrhoeic dermatitis may be evident on the cheeks and chin during the initial stages of beard growth.

The most common clinical form on the trunk is petaloid (the patches are petal-shaped). This is most frequent in males in the anterior and posterior mediothoracic sites. The patches will sometimes converge to form circinate figures (Figs. 2.17 and 2.18). The pityriasiform variant, a generalized erythemato-desquamative eruption similar to pityriasis rosea, is rarer. This variant is only mildly pruriginous, and resolves spontaneously, albeit more slowly than pityriasis rosea.

In the folds (axillae, groin, submammary region, ano-genital region, umbilicus) the dermatitis presents as an intense, diffuse erythema with clearcut margins and oily scales. There may be ragades and secondary infections; intense exudation appears with sweating or in cases of inadequate treatment. The external genitals are affected in both sexes and may also feature chronic scabbing or psoriasiforme lesions.

The severity and course of seborrhoeic dermatitis are very variable, although the disease always has a tendency to become chronic and recurrent. Bacterial complications and a superimposed contact dermatitis are fairly frequent. Erythrodermia is exceptional. In the course of atopic dermatitis or psoriasis, seborrhoeic dermatitis acquires morphological aspects that are difficult to define.

2.7.4 Diagnosis

The diagnosis is generally easy. In severe, generalized cases HIV infection should be excluded. Differential diagnosis must be made with psoriasis, especially of the scalp (the psoriasis patches are often thickened, bright red and with mycaceous scales). At the level of the scalp, pediculosis with pyodermitis can mimic a seborrhoeic dermatitis. In pityriasis rosea the



Fig. 2.15 Seborrhoeic dermatitis



Fig. 2.16 Seborrhoeic dermatitis

lesions are more diffuse and there is the "mother patch". In the folds, the complaint must be differentiated from mycotic or eczematous intertrigo. The follicular form on the trunk must be differentiated from Darier's disease, in which the papules are brown, dome-shaped and confluent, forming bunches. Histology is in any case diagnostic.

2.7.5 Treatment

First of all, it is important to convince the patient that there is no definitive treatment and so the complaint will require regular treatment for years.

Dandruff is treated with the regular, frequent use of anti-*Malassezia* yeast medicated shampoos, with a selenium sulphide, zinc pyrithione, ketoconazole, and tar base. In severe cases 5% salicylic acid ointment can be applied. In acute forms of seborrhoeic dermatitis of the face and trunk, weak steroids (hydrocortisone 0.5%), combined with sulphur (0.5%) are beneficial. Ketoconazole 2% cream is a logical treatment choice, and can be combined with steroids. Tacrolimus and pimecrolimus are also valid, especially for facial forms. Other topical drugs that can yield good results include metronidazole, ciclopiroxolamine, benzoyl peroxide, and 5% lithium succinate. Forms resistant to topical treatment alone can benefit from a course of UVB therapy; in these cases oral itraconazole (100 mg daily for up to 21 days), and oral terbinafine can also be useful. To treat affected skin folds, symptomatic treatment of intertrigo can help.

2.8 Microbial Eczema

This is a controversial clinical entity that is not acknowledged by all authors. However, it is possible to observe occasional cases in which a primitive bacterial or viral skin invasion is followed by eczematous manifestations that then heal when the primitive infection has resolved. For instance,



Fig. 2.17 Petaloid lesions of seborrhoeic dermatitis

a good example is eczema that occasionally develops around lesions of molluscum contagiosum, without these having been traumatized, and resolves when the infection is over. Another example is eczema that appears around infected wounds and resolves with antibiotic treatment only.

Microbial eczema must be differentiated from infected eczema in which the eczematous infection caused by other causes is complicated by a secondary bacterial or viral invasion even if, in practice, these two conditions can coexist, making differential diagnosis very difficult. A further complication is the fact that the microbial flora present on an eczematous lesion can be linked to a simple colonization rather than to infection.

The histological picture of microbial eczema is that of a subacute eczema, with spongiosis, acanthosis, and hyperkeratosis and parakeratosis; there is an evident perivasal infiltrate in the derma, with polymorphonuclear leukocytes and lymphocytes that invade the epidermis. The pathogenic mechanism is unknown, although it is likely that bacterial antigens can promote a cytotoxic reaction at the skin level.

The clinical picture features intense erythema with microvesicles, that develop around wounds or ulcers, or on moist skin (folds, interdigital spaces on the feet) (Figs. 2.19 and 2.20). Staphylococci or streptococci can be cultured, and the lesions respond to antibiotic treatment [105], which will consist of local and systemic antibiotics and local antiseptics (potassium permanganate soaks for 2–3 days if there are exudative lesions).



Fig. 2.18 Petaloid lesions of seborrhoeic dermatitis



Fig. 2.19 Microbial eczema



Fig. 2.20 Microbial eczema

2.9 Dermatophytide

This is a distant reaction starting from a dermatophyte infection [106–108]. The diagnosis is supported by the absence of fungi at the dermatophytide lesions, and by their resolution when the starting lesion is treated. Dermatophytide, therefore, is a distant, secondary aseptic lesion. The condition is rarely observed. Various clinical patterns have been described. The classic example is a symmetrical eczematous area on the sides of the fingers of the hands, starting from a tinea pedis.

A dermatophytide onset is more likely in cases of inflammatory dermatophytes, such as *Trichophyton mentagrophytes* of the zoophilic type [108]. Candidiasis (levuride) is a comparable allergic reaction to a yeast-induced infection.

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